

Regulatory Compliance & Graded Safety Framework for Modular Tokamak Battery Arrays

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1 Executive Summary

This document outlines the safety and regulatory alignment of the 32-unit Modular Tokamak Battery Architecture with International Atomic Energy Agency (IAEA) Safety Standards Series, applying a graded approach tailored to magnetic fusion.

2 Key Safety Pillars

1. **Inherent Self-Limiting Physics:** Fusion reactions cannot experience runaway melt-downs; any plasma instability results in immediate termination of the burn.
2. **Active Disruption Mitigation:** The 36-sector DEC matrix provides active Lenz-law velocity damping, stopping thermal shock loads before plasma reaches first-wall components.
3. **Compact Safety Perimeter:** Because nuclear meltdown risks are zero, the plant footprint is reduced from 500 hectares down to a 63-hectare footprint with a 250 m exclusion zone.